Networks Decide for Us: Emergence of Adoption in Homophily-Imputed Social Topologies

Aníbal Olivera Morales[a,*], Jorge Fábrega Lacoa[a]

[a] *Centro de Investigación en Complejidad Social (CICS), Universidad del Desarrollo, Santiago, Chile*

Abstract

The diffusion of collective behaviors is often modeled using synthetic network topologies, ignoring the empirical reality of multidimensional homophily. This paper asks: how do empirically calibrated topologies act as an autonomous causal force in diffusion outcomes? We impute plausible network structures based on homophily parameters from the GSS (2004) onto attitudinal survey respondents (ATP 2014/2016). Simulating a hybrid complex contagion, we compare these empirical topologies against degree-preserving randomized baselines. We identify two structural mechanisms: First, empirical homophily yields a structural premium, reducing the odds of adoption failure by 57% through local echo chambers. Second, topology governs empirical criticality. Homophilic networks yield predictable, continuous phase transitions (Mean-Field universality, $\gamma \approx 1$), whereas randomized topologies collapse into discontinuous regime shifts. Ultimately, network structure acts as an active mechanism that decisively shapes the volume and dynamics of collective behavior.

Keywords: Social Diffusion, Complex Contagion, Homophily, Agent-Based Modeling, Phase Transitions, Structural Premium

1. Introduction

The sudden emergence of collective behaviors—from the rapid spread of technological innovations to large-scale social movements and unpredictable market anomalies (Gladwell, 2002; Farrell, 1998)—remains a central puzzle in the social sciences. Historically, research into collective action has navigated two distinct pathways: individual decision models focusing on rational utility (Oliver, 1993), and collectivity models emphasizing social interaction and threshold-based coordination (Granovetter, 1978; Valente, 1996).

While recent literature has brilliantly bridged these paradigms through models of "complex contagion" (Centola and Macy, 2007), much of this computational work remains tethered to synthetic, theoretical network topologies (e.g., Small World or random graphs). As critics have noted, this "structural reductionism" often treats agents as cognition-free relay stations (Goldberg, 2021), ignoring the empirical reality that human networks are profoundly shaped by multidimensional homophily—the tendency of similar individuals to associate and cluster (McPherson et al., 2001; McPherson and Smith, 2019). Furthermore, well-known generic models may not fit well to the social problem at hand, as real-world social networks jointly exhibit high clustering, assortativity, short path lengths, and right-skewed degree distributions (Talaga and Nowak, 2020).

To bridge this gap, this paper asks: *Conditional on fixed individual propensities, how do empirically calibrated homophilic topologies act as an autonomous causal force—effectively "deciding for us"—by simultaneously generating a "structural premium" in overall diffusion and dictating the fundamental physics of the phase transition?* (see Section S2 in Supplementary Material for detailed ERGM imputation).

We decompose this macro-sociological inquiry into two distinct structural mechanisms:

1. **The Structural Premium (Volume):** How does empirical homophily systematically amplify the macroscopic scale of diffusion compared to randomized baselines?
2. **Empirical Criticality (Dynamics):** How does the disruption of empirical homophily alter the nature of social tipping points, shifting the system from a predictable, continuous phase transition to a discontinuous, unpredictable regime shift?

2. Theoretical Background

Classic threshold models (Granovetter, 1978) demonstrated that nearly identical groups could produce opposite collective outcomes based entirely on slight variations in threshold distributions. Valente (1996) contextualized this by anchoring thresholds within personal social networks. Building on this, Centola and Macy (2007) showed that behaviors requiring social reinforcement (complex contagions) fail in networks optimized for simple information

*Corresponding author
Email address: an.oliveram@udd.cl (Aníbal Olivera Morales)

spread, introducing the concept of structural phase transitions in collective behavior.

More recently, percolation-based models (Tur et al., 2018, 2024) have incorporated intrinsic utility alongside global influence parameters. However, influence in reality is rarely uniform. When it comes to peer influence, dyads that are socially proximate tend to exert far greater influence over one another (Centola, 2011). Addressing this demands models equipped with plausible network topologies, empirically derived utility distributions, and tie-specific, homophily-weighted influence parameters.

3. Methods: Hybrid Agent-Based Modeling and Network Imputation

We model diffusion as a hybrid process integrating Rational Choice and Selective Social Influence, simulated via the `netdiffuseR` package (Vega Yon et al., 2025).

3.1. Rational Choice (Intrinsic Utility)

An innovation possesses an Intrinsic Utility Level (IUL) defined as $\Gamma \in [0, 1]$. Individuals are characterized by a Minimum Utility Requirement (MUR), $q_i \in [0, 1]$, representing their innate aversion to the innovation. An agent adopts independently if the utility exceeds their aversion:

$$q_i \leq \Gamma \implies a_i = 1 \tag{1}$$

3.2. Selective Social Influence

If intrinsic adoption fails, agents may still adopt via peer contagion. We adapt the classic network threshold formulation, calculating exposure as:

$$\tilde{E}_i \equiv \frac{\sum_{j \neq i} \mathbf{X}_{ij} \tilde{a}_j}{\sum_{j \neq i} \mathbf{X}_{ij}} \tag{2}$$

where adoption via influence occurs if $\tau_i \leq \tilde{E}_i$ (τ_i is the individual network threshold). Crucially, $\tilde{a}_j$ accounts only for infected contacts who are *influential*. Influence is constrained by a Maximum Social Proximity (MSP) parameter, h, compared against the empirical social distance in Blau space, d_{ij}:

$$\tilde{a}_j = 1 \iff a_j = 1 \wedge U_{ij} \leq \sigma(h - d_{ij}) \tag{3}$$

where $U_{ij} \sim \text{Unif}(0, 1)$ and $\sigma(z) = 1/(1 + \exp(-z/T))$.

3.3. Network Imputation Pipeline

To overcome synthetic limitations, we employ the imputation pipeline established by McPherson and Smith (2019). Using Exponential Random Graph Models (ERGMs), we extract homophily strength parameters across demographic dimensions (Age, Sex, Education, Race, Religion) from the representative ego-centric General Social Survey (Smith et al., 2014). As detailed in Section S2 of the Supplementary Material, the regression parameters indicate that distance strictly constrains interaction probability—particularly regarding race and religion differences.

We impute these structures onto $N = 1000$ respondents from the American Trends Panel (ATP). This generates “plausible" topologies that respect empirical clustering while assigning realistic behavioral propensities (e.g., “Openness to Innovation" and “Collective Action" scores) derived directly from ATP items.

4. Results

We explored the parameter space utilizing 41 levels of IUL (Γ) and 13 levels of MSP (h), running $\approx 4 \times 10^6$ simulations across Plausible (GSS) and Degree-Preserving Randomized topologies. The simulations follow a hybrid complex contagion logic using stochastic, asynchronous exposure updates to circumvent artifactual lock-ins (detailed ODD protocol available in Section S1 of the Supplementary Material).

4.1. Statistical Analysis: Big Additive Models (BAM)

To isolate the marginal effect of the empirical structure across all simulations, we specify Big Additive Models (BAMs). We estimate three dependent variables: *Rational*, *Social*, and *Total Adoption*. The models use a binomial family with a logit link function, incorporating thin-plate regression splines $s(\text{IUL}, \text{MSP})$ to capture non-linear interactions across contagion types. We control for the network's threshold distribution (μ_τ, σ_τ), initial seeding strategy, and network type.

4.2. The Structural Premium: Volume of Adoption

To isolate the marginal effect of the empirical structure, we utilized Big Additive Models (BAMs) with non-linear smooth terms $s(\text{IUL}, \text{MSP})$.

Table 1 demonstrates the *structural premium* (visualizations of the non-linear interaction surfaces for the regression models are available in Section S4 of the Supplementary Material). Randomizing the network structure significantly depresses overall adoption. Under mathematically equivalent parameter conditions, removing empirical homophily reduces the log-odds of total adoption by -0.850, corresponding to a $\sim 57\%$ drop in adoption probability. Homophily acts as a structural catalyst, allowing social reinforcement to safely build up in local echo chambers and overcome innate resistance.

4.3. Empirical Criticality: The Dynamics of Tipping Points

Beyond volume, the network topology dictates *how* adoption occurs. Figure 1 maps the adoption regimes and the structural phase transitions.

In empirical (homophilic) networks, social change manifests as a predictable, continuous phase transition. We observe distinct “Critical Slowing Down" (divergence in

Table 1: BAM Regression Results: Effect of Network Structure and Initial Seeding.

	Rational Ch.		Social Infl.		Total	
Coefficients	Estimate	OR	Estimate	OR	Estimate	OR
(Intercept)	0.428***	–	−0.300***	–	6.534***	–
Thresholds						
τ_{mean}	−0.185***	–	−5.130***	–	−6.944***	–
τ_{sd}	0.032	–	4.210***	–	5.942***	–
Seed (Ref: Random)						
Central	−0.054***	0.947	0.032***	1.033	0.120***	1.127
Closeness	−0.052***	0.949	0.033***	1.034	0.119***	1.126
Eigenvector	−0.051***	0.950	0.032***	1.033	0.118***	1.125
Marginal	0.027***	1.027	−0.020***	0.980	−0.067***	0.935
Topology (Ref: Plausible Network)						
Deg.-Preserving Randomized	−0.114***	0.892	−0.426***	0.653	**−0.850***	0.427
Smooths (all $p < 0.001$)	edf	χ^2	edf	χ^2	edf	χ^2
s(IUL, MSP) : Innovation	9.00	522,837	9.00	420,575	9.00	357,704
s(IUL, MSP) : Collective Action	9.00	581,944	9.00	424,819	9.00	405,514
Deviance Explained	96.9%		71.1%		82.2%	

***$p < 0.001$. Standard errors omitted. OR = Odds Ratio ($\exp(\beta)$). edf = Effective Degrees of Freedom. Observations (N) = $4,093,440$.

simulation relaxation times) along the critical boundary. Applying the Fluctuation-Dissipation Theorem ($\chi = \text{Var}(\Phi)$), we recover an empirical susceptibility exponent of $\gamma \approx 0.977$ (Table 2). This constitutes rigorous proof that the homophilic network perfectly obeys Mean-Field universal scaling laws, providing a “sociological viscosity" that governs the cascade.

Conversely, in the GSS-ER random network, this predictability completely collapses. The γ exponent diverges to unphysical levels ($\approx 3.7-4.6$), and critical slowing down is suppressed. The system behaves like a discontinuous, first-order regime shift: it either completely fails or violently explodes into mass adoption with zero statistical warning.

Table 2: Susceptibility Exponents (γ) across Topologies.

IUL (Γ)	**Plausible (GSS) γ**	**Randomized γ**
0.200	0.977	3.709
0.225	0.892	3.667
0.250	0.892	3.667
0.275	1.539	4.666

Mean-Field prediction: $\gamma = 1.0$. Plausible networks exhibit continuous, second-order scaling. Randomized networks exhibit unphysical discontinuity.

5. Discussion and Conclusion

Our results provide robust computational and mathematical evidence that “networks decide for us." Conditional on fixed individual-level propensities, the topological arrangement itself acts as an autonomous causal mechanism.

First, empirical homophily yields a systematic *structural premium*, proving that relational clustering facilitates behavioral cascades that randomized networks cannot sustain. Second, this topology actively reshapes the *physics* of the diffusion process. Homophily ensures that social tipping points occur as predictable, structured avalanches (continuous phase transitions). Stripping a society of its homophilic structure does not just make adoption harder; it makes it volatile, transforming gradual cascades into unpredictable, sudden explosions.

Ultimately, macroscopic collective outcomes cannot be reduced to the aggregate of individual preferences; they are profoundly, and mathematically, dictated by the structure of our social ties.

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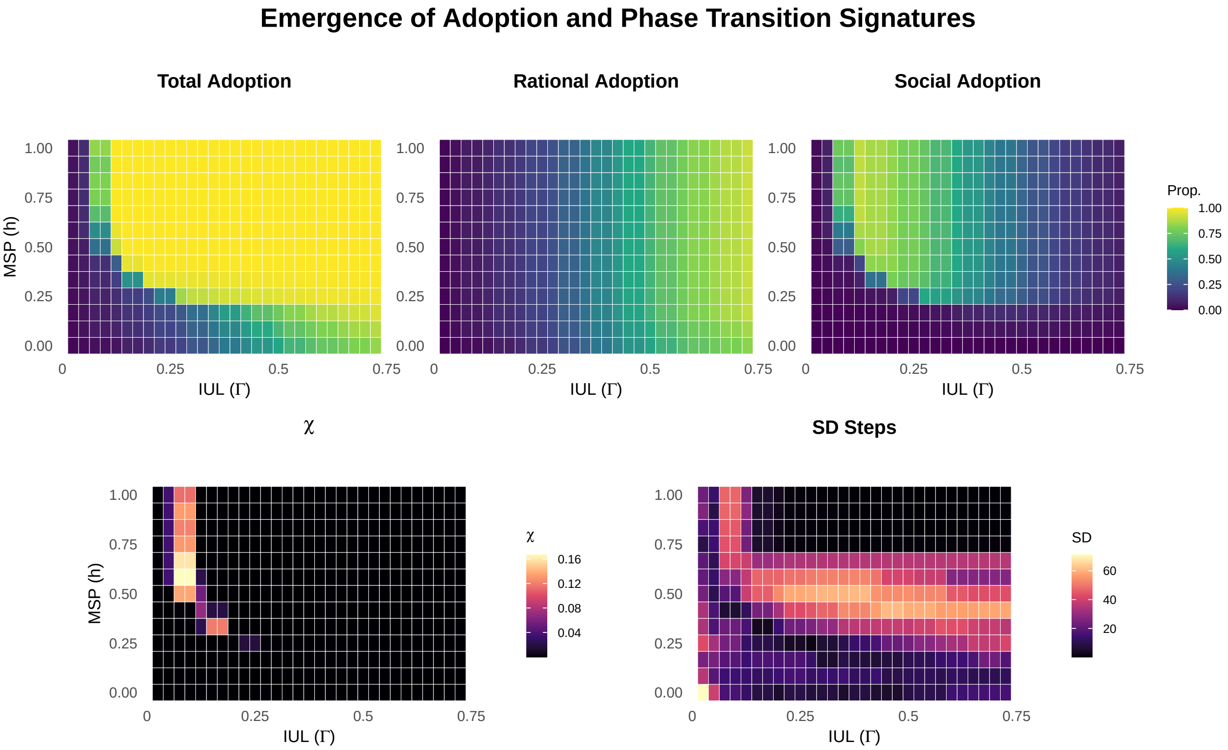

Figure 1: *Adoption Regimes and Phase Transition Signatures.* Top row: Average Total, Rational, and Social Adoption heatmaps. Bottom row (left): Susceptibility $\chi = \text{Var}(\Phi)$ via the Fluctuation-Dissipation Theorem, with the phase boundary visible as a ridge of high variance. Bottom row (right): Standard Deviation of simulation relaxation time; its divergence is the signature of Critical Slowing Down. Simulations based on a GSS plausible network, $\tau_{sd} = 0.12$, and random seeding (see Section S5 for equivalent signatures under Erdős-Rényi topologies).

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